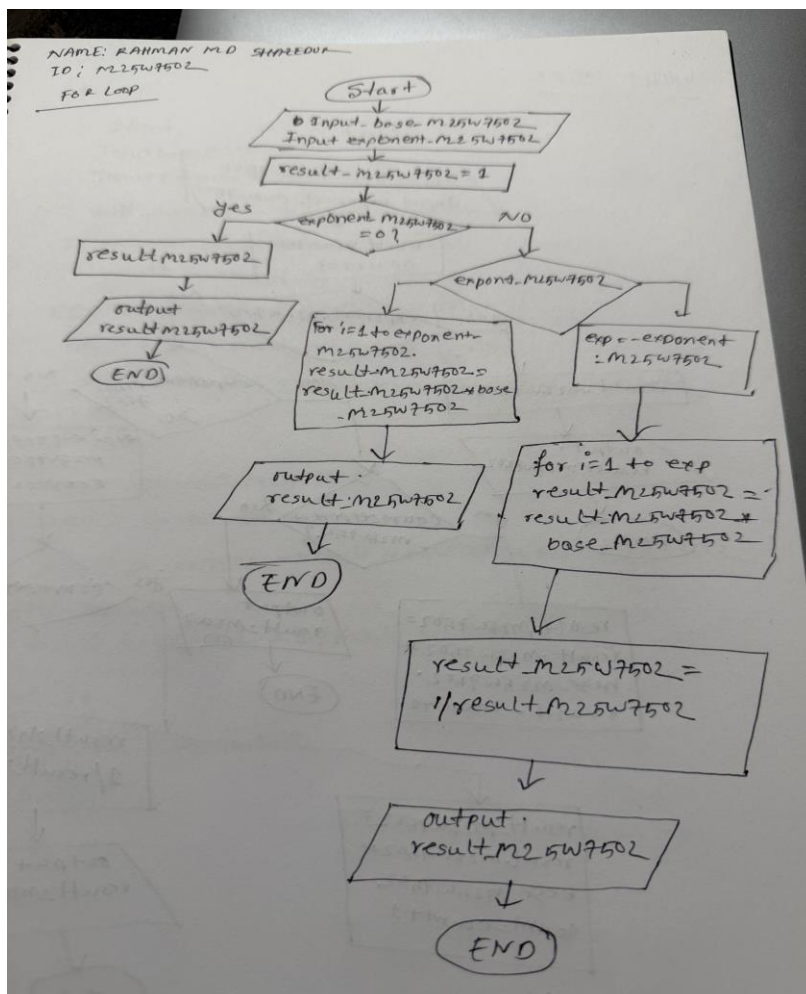


Algorithms Assignment 03

Full Name: RAHMAN MD SHAZEDUER

ID: M25W7502

1. Flowcharts Using for loop



2. Pseudocode

```
FOR Loop:
  Start
  Input base_m25w7502
  Input exponent_m25w7502
  result_m25w7502 = 1
  IF exponent_m25w7502 == 0 THEN
    result_m25w7502 = 1
  ELSE IF exponent_m25w7502 > 0 THEN
    FOR i = 1 TO exponent_m25w7502 DO
      result_m25w7502 = result_m25w7502 * base_m25w7502
    END FOR
  ELSE
    temp = exponent_m25w7502
    FOR i = 1 TO temp DO
      result_m25w7502 = result_m25w7502 / base_m25w7502
    END FOR
    result_m25w7502 = 1 / result_m25w7502
  END IF
  PRINT result_m25w7502
END
```

3. Program:

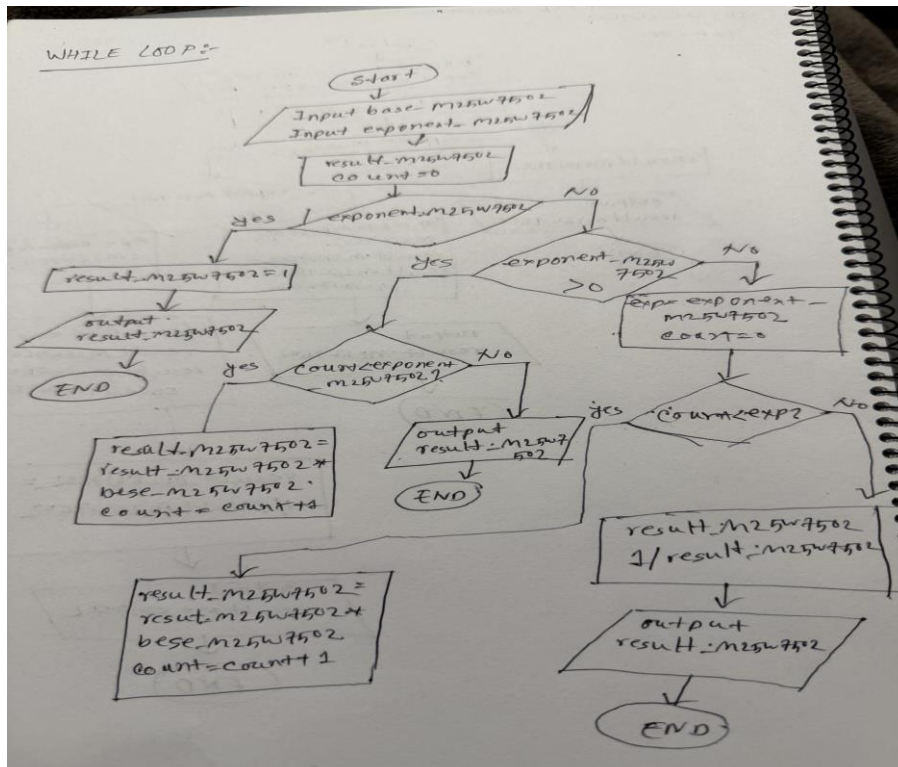
```
1 # =====
2 # Assignment: Exponentiation using Iteration
3 # Name: Rahman Md Shazedur
4 # Student ID: M25W7502
5 # =====
6
7 # Input
8 base_shazedur_M25W7502 = int(input("Enter base: "))
9 exponent_shazedur_M25W7502 = int(input("Enter exponent: "))
10
11 # Initial value
12 result_shazedur_M25W7502 = 1
13
14 # Logic
15 if exponent_shazedur_M25W7502 == 0:
16     result_shazedur_M25W7502 = 1
17
18 elif exponent_shazedur_M25W7502 > 0:
19     for i_shazedur_M25W7502 in range(0, exponent_shazedur_M25W7502):
20         result_shazedur_M25W7502 = result_shazedur_M25W7502 * base_shazedur_M25W7502
21
22 else:
23     temp_exponent = -exponent_shazedur_M25W7502
24     for i_shazedur_M25W7502 in range(0, temp_exponent):
25         result_shazedur_M25W7502 = result_shazedur_M25W7502 * base_shazedur_M25W7502
26     result_shazedur_M25W7502 = 1 / result_shazedur_M25W7502
27
28 # Output
29 print("Result = ", result_shazedur_M25W7502)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

Python

```
/usr/local/bin/python3 "/Users/shazedurrahman/Desktop/ASSIGNMENT/Algorithms /Assignment 03 /power_for_loop_M25W7502.py"
(base) shazedurrahman@Shazedurs-MacBook-Pro Assignment 03 % /usr/local/bin/python3 "/Users/shazedurrahman/Desktop/ASSIGNMENT/Algorithms /Assignment 03 /power_for_loop_M25W7502.py"
Enter base: 5
Enter exponent: 7
Result = 78125
(base) shazedurrahman@Shazedurs-MacBook-Pro Assignment 03 %
```

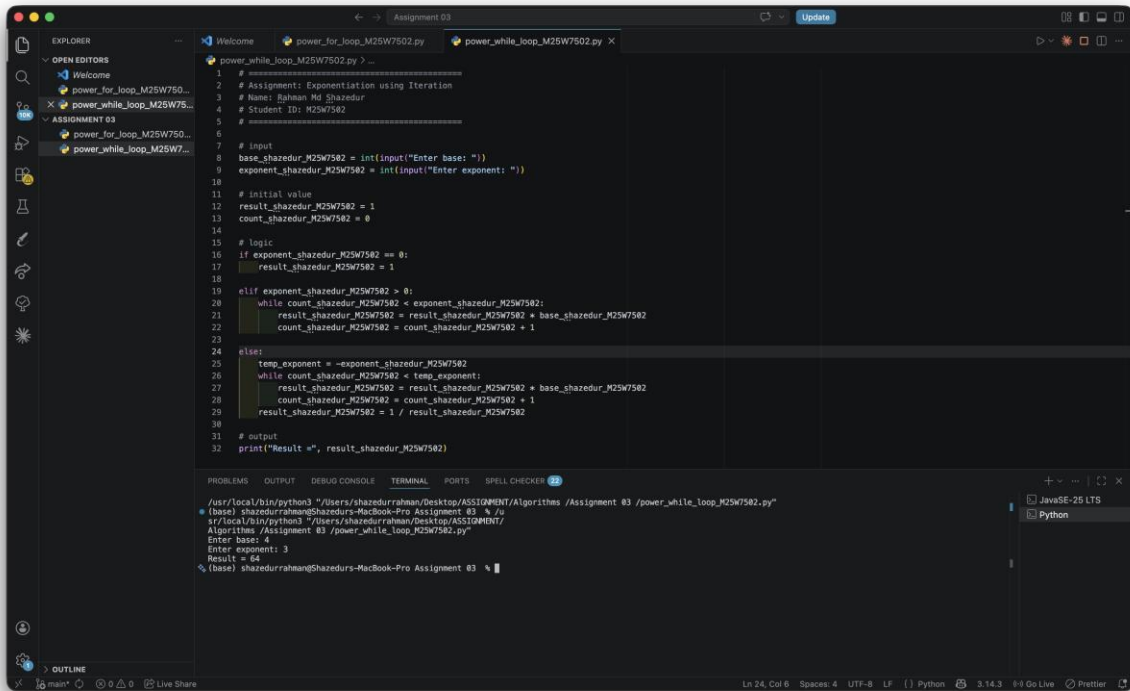
Using while loop



2. Pseudocode

```
WHILE LOOP:-  
  
Start  
Input base-M25W7502  
Input exponent-M25W7502  
result-M25W7502 = 1  
count = 0  
  
IF exponent-M25W7502 <= 0 THEN  
    result-M25W7502 = 1  
ELSE IF exponent-M25W7502 > 0 THEN  
    WHILE count < exponent-M25W7502 DO  
        result-M25W7502 = result-M25W7502 * base-M25W7502  
        count = count + 1  
    END WHILE  
ELSE  
    temp = exponent-M25W7502  
    WHILE count < temp DO  
        result-M25W7502 = result-M25W7502 * base-M25W7502  
        count = count + 1  
    END WHILE  
result-M25W7502 = 1 / result-M25W7502  
END IF  
PRINT result-M25W7502  
END IF  
PRINT result-M25W7502  
END
```

Program:



Code In Text Format:

Using for loop

Assignment: Exponentiation using Iteration

Name: Rahman Md Shazedur

Student ID: M25W7502

=====

input

base_shazedur_M25W7502 = int(input("Enter base: "))

```
exponent_shazedur_M25W7502 = int(input("Enter exponent: "))

# initial value
result_shazedur_M25W7502 = 1

# logic
if exponent_shazedur_M25W7502 == 0:
    result_shazedur_M25W7502 = 1

elif exponent_shazedur_M25W7502 > 0:
    for i_shazedur_M25W7502 in range(0, exponent_shazedur_M25W7502):
        result_shazedur_M25W7502 = result_shazedur_M25W7502 *
        base_shazedur_M25W7502

    else:
        temp_exponent = -exponent_shazedur_M25W7502
        for i_shazedur_M25W7502 in range(0, temp_exponent):
            result_shazedur_M25W7502 = result_shazedur_M25W7502 *
            base_shazedur_M25W7502

        result_shazedur_M25W7502 = 1 / result_shazedur_M25W7502

# output
print("Result =", result_shazedur_M25W7502)
```

Using while loop

Assignment: Exponentiation using Iteration

Name: Rahman Md Shazedur

Student ID: M25W7502

=====

input

base_shazedur_M25W7502 = int(input("Enter base: "))

exponent_shazedur_M25W7502 = int(input("Enter exponent: "))

initial value

result_shazedur_M25W7502 = 1

count_shazedur_M25W7502 = 0

logic

if exponent_shazedur_M25W7502 == 0:

result_shazedur_M25W7502 = 1

elif exponent_shazedur_M25W7502 > 0:

while count_shazedur_M25W7502 < exponent_shazedur_M25W7502:

result_shazedur_M25W7502 = result_shazedur_M25W7502 *

base_shazedur_M25W7502

count_shazedur_M25W7502 = count_shazedur_M25W7502 + 1

else:

temp_exponent = -exponent_shazedur_M25W7502

while count_shazedur_M25W7502 < temp_exponent:

result_shazedur_M25W7502 = result_shazedur_M25W7502 *
base_shazedur_M25W7502

count_shazedur_M25W7502 = count_shazedur_M25W7502 + 1

result_shazedur_M25W7502 = 1 / result_shazedur_M25W7502

output

print("Result =", result_shazedur_M25W7502)